

12 (a) loss / reduction in power / energy / voltage/ amplitude (of the signal) B1 [1]

(b) (i) attenuation = $125 \times 7 = 875$ dB A1 [1]

(ii) 20 amplifiers
gain = $20 \times 43 = 860$ dB A1 [1]

(c) gain = $10 \lg(P_1/P_2)$ C1
overall gain = -15 dB / attenuation is 15 dB C1
 $-15 = 10 \lg(P / 450)$
 $P = 14$ mW A1 [3]

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13 (a) switch: tuning coil: (r.f.) amplifier: demodulator: